

Shades of sustainability – is there room for a new sustainability label in fruit production?

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Abstract

We are taking advantage of the development of a new sustainability label in German fruit production that is aimed at the end consumer as a case study to conduct an ex-ante assessment of consumer and producer acceptance of a new label. We present here the first steps toward a decision analysis project, which includes the evaluation of two surveys and the design of a graphical influence model. The results of the two surveys summarize the specific conditions under which a new label can be successful from a producer and consumer perspective. The model visualizes the influence pathways opened up by the introduction of the label. We attempt to predict the success of such a label based on the theory of innovation diffusion by examining niches in fruit production that could influence the mainstream of fruit growing and marketing through a window of opportunity. Labels are a proven means to accomplish several objectives in the field of sustainable resource management. Among these, we explored the role of a label in satisfying consumer information needs and in promoting innovation in a production system in order to forecast the system change potential of the label. We further assume that the impact of the label on the system largely depends on the acceptance of the innovative label. The acceptance of an innovation such as a new label can be decisively promoted if it corresponds to the habits, norms and needs of customers. We therefore asked 1000 German consumers what developments they would welcome in fruit production and where they see the greatest need for action. In order to be able to keep the sustainability promises of the label, a change in production processes is necessary. To do this, the criteria of the label must be incorporated into the decisions of the fruit growers. We therefore asked the fruit growers what changes they would be willing to make and what means they consider suitable for making fruit growing more sustainable. The results of the surveys were used to compute a forecast of the diffusion of the innovative label. The results will be incorporated into a conceptual decision support model that will provide recommendations for the further development of the label.

Keywords: consumer attitudes, decision support, adoption of innovations

INTRODUCTION

In August 2022, the ISHS Sustainability Declaration of Toronto 2002 will be exactly 20 years old. The demands contained therein offer a holistic, participatory and comprehensive approach to reducing negative social and ecological impacts of horticulture (Bertschinger et al., 2004). These approaches are still relevant today. Fruit growing, as an important sector of horticulture, contributes to climate change with greenhouse gas emissions estimated at 0.88, 0.64 and 0.46 kg CO₂-eq per kg of fruit consumed for plums, cherries and apples, respectively (Svanes and Johnsen, 2019). Poorly managed fruit farms can pose a risk to human health – especially to farm workers – through the use of chemical pesticides (Calvert et al., 2008). To be truly sustainable, agricultural systems should provide sufficient high quality yields over the long-term. They should also be profitable, protect the environment, conserve resources and be socially responsible (Reganold et al., 1990). Fruit growing offers potential for improvement in terms of its sustainability – especially in those areas that do not relate

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exclusively to ecological indicators. If one takes the Toronto Declaration seriously, the areas of social concerns and economic sustainability should be given equal consideration, and a more holistic, systemic view of the issue should be the order of the day. We see a need for further action, especially with regard to a systemic approach, and see our contribution in this paper, as it looks at as many areas of sustainability as possible from two different perspectives and includes important actors in the analysis.

Today's fruit growing in Germany, and also in large parts of the European and other world regions, represents a highly efficient, knowledge-based, specialized production system that has been optimized down to the individual farm processes. Any changes in production methods pose high risks for fruit farms. For the transformation of fruit growing toward more sustainable production, which might require considerable, possibly disruptive changes, appropriate incentives are needed in order to make the risk calculable for the farms. The nature of these incentives, whether financial, novel marketing options or targeted information, may depend on the individual situation of farm and farmer and need to be explored before starting a change. Compared to other European countries, Germany does not yet have a sustainability label that is visible to consumers. Labeling is used to balance the asymmetry of information between producers and consumers and to incentivise socially desirable purchasing behavior. At the same time, labels that signal sustainable production increase consumers' willingness to pay price premiums of up to approximately 30% (Li and Kallas, 2021). Labeling thus offers fruit farms a means to distinguish their products from non-labeled products and the opportunity to secure a premium price.

The only labels certified by a third party that are currently used in consumer communications to visualize efforts to increase sustainability in fruit production are those for certified organic agriculture. The history of organic labels implies that a complete transformation of the production system may be necessary to meet organic sustainability standards (Falconer, 2000; Zikeli et al., 2014). Furthermore, after a transition period, there must be a high probability that higher product revenues will be achieved. In times of economic crises that affect consumers' purchasing power, high produce prices represent a risk for certified sustainable products. However, such crises may also bring opportunities, since such systems are likely to be less dependent than conventional systems on external inputs whose prices may also rise sharply in times of crisis. Labels are used to show compliance with ecological sustainability standards and enhance the willingness to pay on the part of the consumer. Although the organic label promises a premium price for the product, many farmers hesitate to convert to organic production, and some even give up organic production. This indicates that conversion decisions are much more likely to be based on farmers' individual personal motivations and values than on a premium price (Darnhofer et al., 2005), while the abandonment of organic certification is often caused by the perception of excessive regulation and disappointed expectations of increased revenue (Flaten et al., 2010). The barriers to conversion to organic agriculture and the reasons for leaving organic agriculture account for the rather moderate growth in the extent of organically farmed area and thus contribute to a growing gap between consumer demand and a stagnating supply of products from sustainable production.

In this paper, we explore the question of whether the desire for a new sustainability label is comparable to the desire for products from sustainable production. We try to assess whether it could be a good idea to introduce a sustainability label in Germany before its introduction ex-ante. We would like to work on this question with the steps of a decision analysis and present here first results. For fruit farmers it will be a decision whether to choose a label for their products and invest in necessary changes of their farming system. A growing number of conventional or integrated fruit farms have begun adopting selected sustainability measures, leading to diversification of fruit production practices. Many fruit growers who market a high proportion of their harvest directly and are in direct contact with customers see an important unique selling point in emphasizing the sustainability of their production. Several retail chains also appeal to sustainability-oriented customers by developing their own sustainability labels and trademarks. Given the dominant position of such retailers in the market, this development will become increasingly important.

Given these dynamic developments in the field of sustainability labeling, we set out to find a “window of opportunity” (Geels and Schot, 2007) for a label that considers all areas of sustainability and appeals to fruit growers and consumers alike. We used Rogers’ diffusion theory and applied it to our case by attempting to capture all relevant stakeholders and impacts that would promote or hinder adoption of the proposed label and visualize them as actors and influences in an impact model. We integrated our findings to create a comprehensive conceptual model for label dissemination. . We show key results of two surveys aimed at collecting information from key stakeholder groups to feed the model. The result is a decision framework into which calculations can be embedded to help decide whether it makes sense for farms to be certified to the standards of an innovative sustainability label. We expect that the probability of success of an innovative sustainability label for fruit may be high if:

- The demands of the consumers completely correspond to the willingness of the producers to change their behavior or are largely in line with it;
- The possible changes in the product characteristics of the labeled product (e.g. product color, shape, price) that results from more sustainable farm management are within the tolerance range of the consumers;
- The relative advantage of the innovation is clearly visible for both groups.

MATERIALS AND METHODS

Theoretical framework

The Trilogy of the diffusion of new goods or standards developed by Schumpeter recognizes three major stages: invention, innovation and diffusion (Stoneman and Diederer, 1994). In Schumpeter’s view, progress requires an entrepreneurial force that drives the development of the economic landscape and shapes it through invention (creation of a new idea), innovation (concrete application of the new idea) and diffusion (the spread of the innovation across its potential market) (Schumpeter and Backhaus, 2003). The third step, the diffusion, will be at the core of this contribution, with an innovative sustainability label as corresponding innovation. The diffusion of innovations requires that the targeted market actors become adopters by starting to make use of the innovation. We base our work on the diffusion of innovation concept by Everett Rogers (Rogers, 1962). Rogers defines diffusion as the process by which an innovation is communicated over time among the members of a social system through certain channels. The speed with which an innovation spreads within a group is called the adoption rate (AR). We built a conceptual diffusion model following a conceptual model to predict virus spread (Duggan, 2017). The model describes the spread of a virus as a function of the number of potential infections and the effective per capita contact rate. In Duggan’s approach, the adoption rate (AR) reflects the dynamics of infection. In our adapted concept, AR represents the adoption of innovation. We adapted the model to account for consumers’ and producers’ assessments of innovation characteristics, as well as other innovation drivers and their influences on the process. AR is expected to be influenced by 1) the product characteristics, which according to Rogers consist of five attributes: complexity, relative advantage, compatibility, trialability and observability and 2) the social dynamics related to the adoption of an innovation. With market saturation, the AR is expected to decrease due to a decline in the number of potential adopters. To really allow inferences about adoption processes, this model will be included in an overall model of the entire production system from production to consumption.

Definition of adoption and identification of its drivers

We consider the adoption behavior of two groups of actors – producers and consumers. For each group, we define characteristics that can be used to judge whether an innovation has been adopted or not. For our case – the introduction of a new label – we first define what successful adoption of the label means.

Following Rogers, the two possible outcomes of a decision-making process are adoption or rejection. We consider decisions as “irrevocable allocations of resources” (Howard, 1966).

Based on this definition of a decision the criterion for adoption on the producer's side is the signed contract with the labeling organization and an adaptation of the agricultural production system to the given production guidelines. On the consumer's side, assessing adoption of a new product label is less straightforward, because such a label can never be consumed on its own, but only in conjunction with an actual product. Therefore, both the expected characteristics of the product and the innovation – the label – need to be taken into account to assess whether adoption occurred. The mere purchase and consumption of the labeled product, while representing an irrevocable allocation of resources, is not sufficient as an indicator, since this process may also take place purely by chance. Accordingly, the labeled product needs at least one feature that distinguishes it from the unlabeled product and whose conscious acceptance can be considered as an indicator of successful adoption. For our case, we chose three characteristics: the consumer accepts a higher price, the consumer accepts deviations from the optical norm (color deviation, stains, deviating size) and the consumer accepts the use of a different shopping environment than the one he is used to.

For the ex-ante assessment of adoption, we could not measure the conclusion of contracts or conscious purchase decisions, so we used two surveys to ask about the willingness to change behavior within the group of German consumers and German fruit producers.

Survey design

Our main research interest was to find out which factors influence the success of the new label for sustainable fruit production in Germany. Success means that a critical mass of consumers and producers adopt the label. How large this critical mass must be still needs to be determined and is outside the scope of this contribution, yet our hypothesis was that the probability of success may be high if a) the demands of the consumers completely correspond to the willingness of the producers to change their behavior or are largely in line with it, b) the possible changes in the product characteristics of the labeled product (e.g. product color, shape, price) that result from more sustainable farm management are within the tolerance range of the consumers, and c) the relative advantage of the innovation is clearly visible for both groups.

In order to assess these attitudes and motivations of the stakeholder groups in Germany with regard to sustainable fruit production and to test our hypothesis, we conducted two surveys using online questionnaires. Both questionnaires were transferred into an online format. The consumer questionnaire was submitted to a contracted market research institute. We determined that the panel should consist of individuals who met certain criteria: the gender ratio should represent the German average, the age of the respondents should be between 18 and 75 years, in order to mainly include people who also buy food for their household, and the respondents should mainly reside in North Rhine-Westphalia, because this is where the introduction of the label is being considered. With these specifications, the institute put together a panel of around 2700 people. The consumer panel survey took place in February 2022 and resulted in 1512 returned questionnaires, of which 1003 could be considered for evaluation.

The producer survey design was tested with fruit growers and official representatives of the nationwide organization and representation of the interests of fruit growers and the chambers of agriculture in each federal state. The online producer survey was sent out in early March 2022 via separate emails to the distribution lists of the working groups in all federal states and was open for three weeks. Since fruit farms are only surveyed every 5 years, we refer to the figures for the number of fruit farms from 2017 (Garmin et al., 2018), where a figure of around 11,000 fruit farms in Germany was calculated. Of these approximately 11,000 farms, we received 232 responses, of which 114 were included in the evaluation.

The questionnaires were structured as mirror images, one from the perspective of the consumers, one from the perspective of the producers. For example, we asked the consumers which special achievements of the producers in the area of sustainability should be made visible through a sustainability label. The producers were asked which of their special achievements in the area of sustainability should be emphasized by a label.

We designed a structured questionnaire to evaluate consumer opinions. We pretested this questionnaire at a public event dedicated to the Sustainable Development Goals on the market square of the city of Bonn on three consecutive days. For this purpose, 50 passers-by were interviewed, supported by a pen-and-paper version of the questionnaire. With regard to the sample size of a pre-test, it is recommended to go up to the level of redundancy. For most occasions, this is already achieved with a sample size of 30 persons (Perneger et al., 2015), so that a sample of 50 persons can be considered robust. The specific results of the pre-test were discarded as they only served to check and improve the comprehensibility of questions and design. Adjustments to the draft were made based on the respondents' reaction to ensure optimal comprehensibility and appropriate structure of the questions. When choosing the questions, we considered both open and closed questions, including point ratings, Likert scales and multiple choice options.

RESULTS

Characteristics of survey respondents

The consumer survey resulted in 1512 responses from the consumer panel, of which 1003 were evaluated as valid contributions. 413 responses had to be excluded due to incompleteness, 96 due to an unanswered quality assurance question. The distributions of age, gender, income, occupation, education and household size are largely comparable to the distributions in the overall German population, although the share of female respondents was slightly higher compared to the average German population (54% female respondents vs. 51% female German population). This situation does not compromise the validity of the findings, but should be considered when interpreting the results. Overall, 96% of the questions were answered by respondents who stated that they were the main person doing all grocery shopping or at least doing half of it, 93% of the respondents said that they consume fruit, while 37% reported eating fruit on a daily basis. These findings imply that the respondents influence the consumption behavior of the household and that fruit is one of the regularly purchased goods.

The producer survey resulted in 232 responses from the mailing list of the German fruit growers association, of which 114 were evaluated as valid contributions. 118 responses had to be excluded due to incompleteness. The distribution of farm sizes and production methods (organic, integrated production, conventional), as well as the proportion of farms from the different federal states, largely corresponded to the distribution in Germany. An overrepresentation of fruit farms from the Hamburg region (9% of the participants in the survey compared to 4.6% of the farms in Germany (Garmin et al., 2018) must be considered for further analysis. Fewer than 10% of survey respondents stated that they had no successor and wanted to close down the business. It can therefore be assumed that a focus on the future is important for most of the participating fruit farms. Overall, 102 persons stated that they were the manager of the farm, and another 7 stated that they were a family member working on the farm. It can be expected that the large majority of respondents are people who are in a position to make key decisions on the farm.

Relative advantage, compatibility, complexity and observability

We illustrate here the results of the relative benefit to both stakeholder groups (Figure 1). The analyses of complexity, compatibility and observability are then carried out in the same way, but cannot be dealt with in detail here. The most important step in assessing the relative advantage for consumers was to identify the content that consumers would find most relevant in a sustainability label. The consumers rated abandoning synthetic herbicides (52%), abandoning synthetic insecticides and fungicides (49%), reducing the amount of packaging (43%), reducing the use of plastic (35%) and reducing food waste (30%) as the 5 most important aspects out of 17 options. On the producer side, most respondents said that making their support for biodiversity visible was most important (53%), followed by generating their own energy (39%), focusing on regional marketing (34%), pursuing their own sustainability strategy (27%), and avoiding food waste (24%). From the perspective of

producers, 41% said that there were already too many labels on the market and that they therefore did not consider another label to be useful.

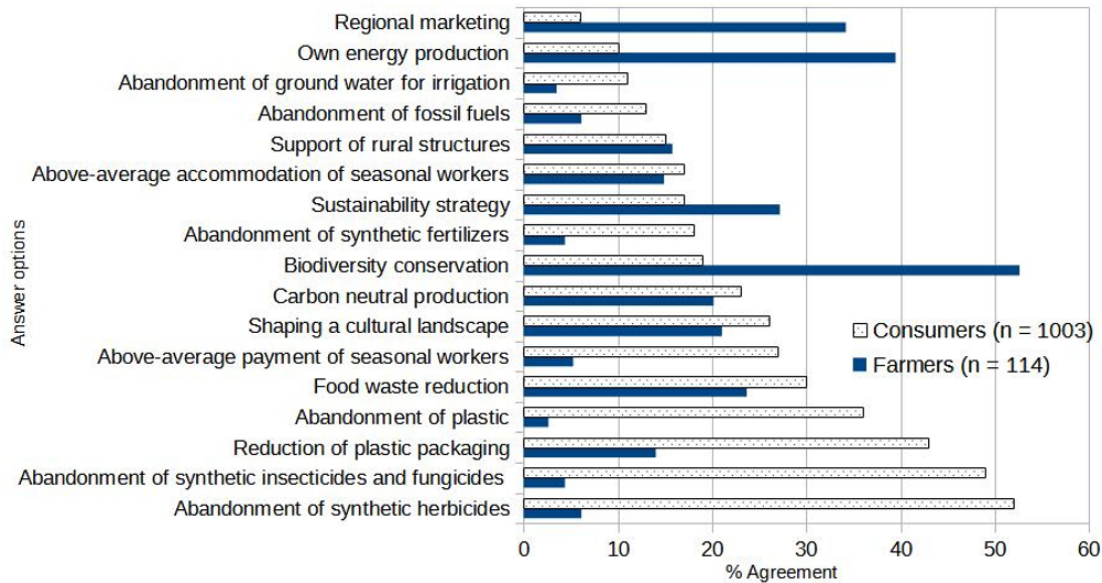


Figure 1. Relevant content for a sustainability label from consumers' and farmers' perspective. Consumers prefer to have the abandonment of chemical agents in fruit growing made more visible through a label, producers would prefer to make their efforts in the preservation of biodiversity, their own energy production and regional marketing more visible.

A slight majority of consumers (54%) agreed with the statement that fruit producers alone will not create a change toward greater sustainability and therefore need to be supported to become more sustainable, while 39% were not sure and 7% disagreed. When asked what they themselves would be willing to do to support this change, consumers said they would specifically buy from sustainable farms (49%), specifically look for a sustainability label (42%) or sign an online petition (36%). 73% of the consumers surveyed saw the greatest influence on sustainability-oriented decision-making behavior on the part of producers in the requirements of wholesalers and retailers, 71% in special financial incentives (subsidies, direct payments, favorable loans, etc.), 70% in a positive attitude toward sustainability on the part of the fruit farmer.

In general, 63% of the consumers and 35% of the producers said that a label can be a useful or a very useful tool to promote sustainability in fruit production. This may highlight a discrepancy between the needs of the two groups. The consumers' need for information is contrasted with the producers' need to make their own efforts for sustainability visible. The consumers' need for information and their conviction of the suitability of a label for this purpose is thus significantly greater than the producers' need to show their efforts. In other words, their confidence in the signal effect of such a label is significantly lower than that of consumers. This difference in attitude toward the usefulness of the label can be explained by the burden distribution associated with such a label. While consumers have the relatively easy task of actively searching for the label and paying a certain premium, many farmers are in a situation where they may have to change their farming system and elaborated habits (Figure 2).

Diffusion model

We built a conceptual diffusion model (Figure 3) following a conceptual model to predict virus spread (Duggan, 2017), which we adapted to our application case. We adapted

the virus model to account for consumers' and producers' assessments of innovation characteristics, as well as other innovation drivers and their influences on the process. Our model assumes that the first target groups for the introduction of a new sustainability label will be producers and consumers, who can be divided into groups with different response behavior on the basis of socio-economic characteristics, personality traits, attitudes and communication behavior. We adapted the basic model to account for consumers' and producers' assessments of innovation characteristics, as well as other innovation drivers and their influences on the process. We have taken into account the various influences that affect the decisions of the two stakeholder groups. These are not completely identical, but have significant overlaps. To classify the categories of influence that affect the decision for or against the adoption of the innovation, we followed Rogers nomenclature and assigned the responses of the stakeholder groups to the areas of relative advantage, compatibility, complexity and observability. At the core of the concept, there is a driving force that comes from societal challenges, norms, past experiences and practices that affect societal actors and promote societal change. Whether these actors, consumers and producers in our case, recognize a sustainability label as a useful tool to tackle the challenges will ultimately determine the success or failure of the label's introduction.

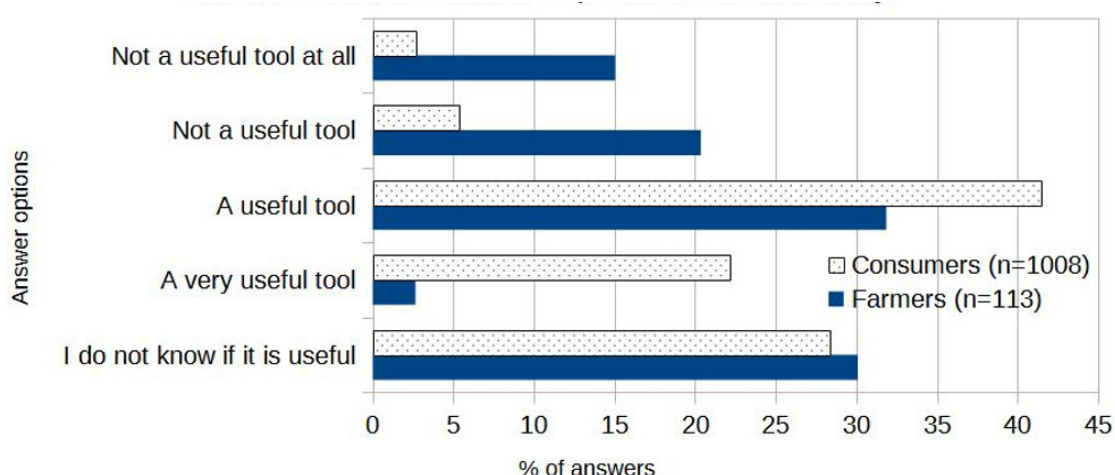


Figure 2. Attitudes of consumers and farmers toward the introduction of a new label. Consumers are more positive than producers about a label as a tool to promote sustainability in fruit growing.

DISCUSSION AND OUTLOOK

We expected that the probability of success of an innovative sustainability label for fruit may be high if it meets our predefined criteria of willingness to behavioral change in both stakeholder groups, tolerance to changes in product characteristics and a clearly visible relative advantage of the label. Based on our results, a new label could indeed be a possible way forward to make certain aspects of sustainability visible. The relative benefit of a new type of label for the consumer could be that, although the aspects ranked as most important for the consumers are already made visible by the organic labels, the other important response options like the reduction of packaging, plastics and food waste, have not yet been addressed in existing certification schemes or are even caused by them. For example, of the GLOBAL GAP standard prohibits marketing of products that do not meet certain visual quality criteria, exacerbating the problem of food waste. For the respondents who support sustainable development in fruit growing and also those who said they actively seek out sustainable fruit farms and sustainability labels, the novel sustainability label can be an advantage in aligning their purchasing decisions with their values.

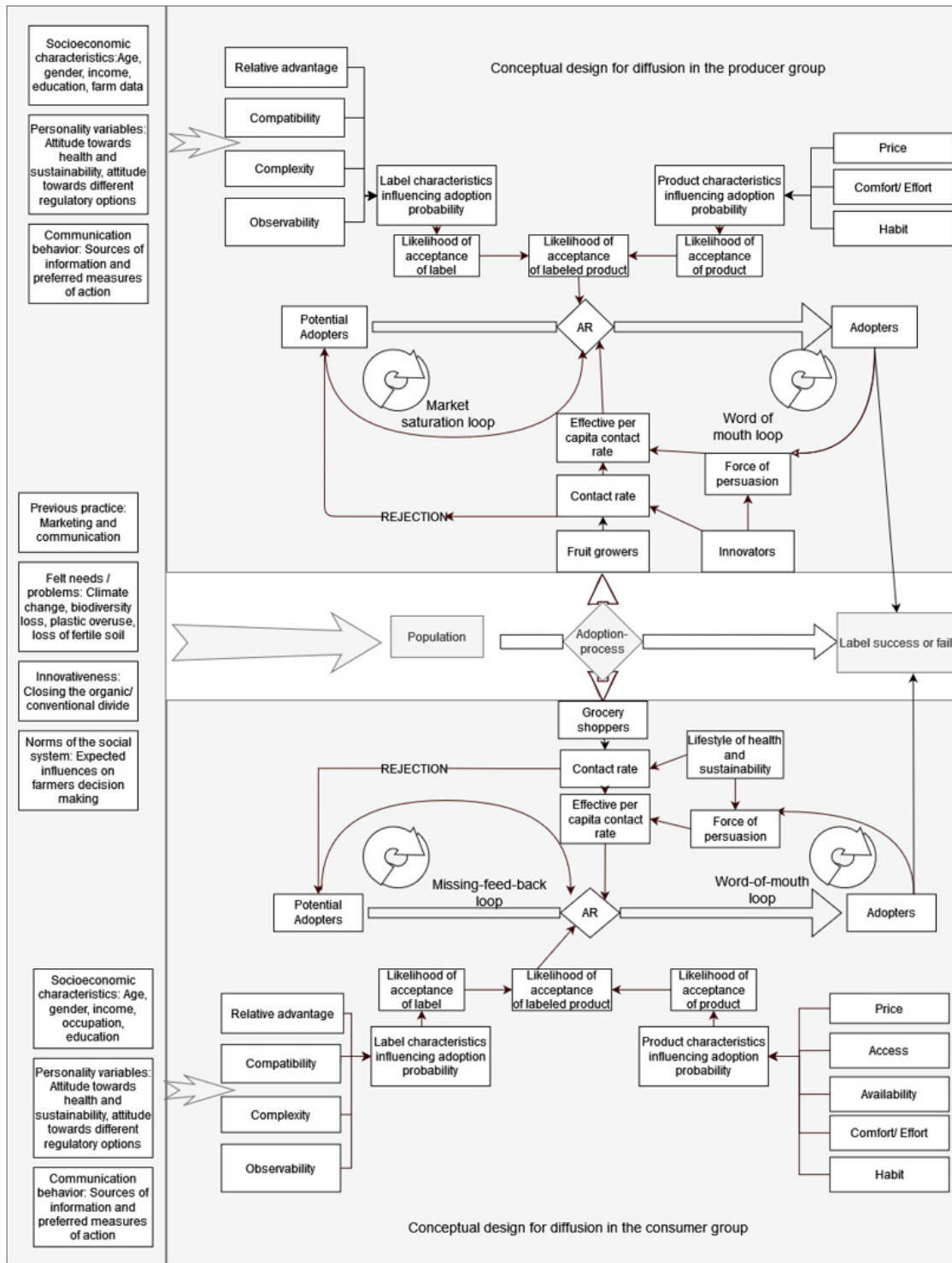


Figure 3. Conceptual diffusion model for a sustainability label in Germany. Consumers and producers have important roles to play in the dissemination of a new sustainability label. While the adoption process is the same, different drivers affect the decision for or against adoption.

It is well known that there is a difference between the response behavior of the respondents and their actual actions when making a purchase (Padel and Foster, 2005).

Purchase decisions can also be influenced differently at the point of sale and are not only made on the basis of the values and attitudes of the respondents but also on the basis of different product characteristics such as the price and lack of visual attractiveness of specific product characteristics. Especially in Germany, price can be a decisive factor, as discount retailers have secured a strong market position there. If we follow Roberts' idea that there are certain individuals who are able to promote change in adopting an innovation as innovators and early adopters, those actors may play a key role in the entire process. Compared to the general population, innovators and early adopters are characterized by a relatively high willingness to take risks, a certain socioeconomic capacity to shape the future, and a willingness to learn and inform themselves (Rogers, 2003). A further analysis could identify such individuals and their potential influencing power.

In particular, the large group of respondents with relatively neutral attitudes toward a new label introduces considerable uncertainty. Considering that about 30% of the surveyed consumers and producers were unsure whether a quality label is suitable to increase sustainability in fruit growing, they could be addressed via suitable information or marketing measures. This could be one of the important leverage points where retail could be involved. Contact with innovators and early adopters can also lead to a more positive attitude.

Some of the challenges are that consumers' information interests align in only a few aspects with the criteria that fruit producers would like to make more visible. The difference in the assessment of the label as a useful tool for promoting sustainability in fruit growing, which is viewed much less critically by consumers than by farmers, may also hinder rapid adoption of a new label. In designing the label, the main task must be to find a middle ground that satisfies the interests of both stakeholder groups without at the same time overburdening interested fruit growers with an excess of certification requirements. Based on Rogers' concept, this seems to be the core task that will determine the success or failure of the label. Building on the results of the surveys and the conceptual model, the next important step would be the development of a mathematical model that allows quantification of the facilitating and hindering influences. To calculate the AR and the probability of success, the R packages "diffusion" (Schaer et al., 2018) and "decisionSupport" (Luedeling et al., 2021) can therefore be used to implement the model with values from the consumer survey. Based on these results of the model, an adapted marketing strategy can be developed. For its creation, all available information is needed, including reliable forecasts. The quality of the marketing concept will be as crucial to the success or failure of the label as the content that the label embodies.

CONCLUSIONS

Our results suggest that there may be a window of opportunity for the introduction of an innovative sustainability label for fruit growing in Germany. The results indicated that the design of the label is likely to play an important role in this. Satisfying the consumers' need for information as well as the fruit growers' need to make their efforts to promote sustainability visible without overburdening the fruit growers were identified as important factors. A clever marketing strategy that would be able to reduce uncertainties on both sides, for consumers and producers, could also be beneficial. Information on the attitude of retailers toward a sustainability label for fruit production would generate additional clarity on the chance of success of such a label. Recent world events, most notably the ripple effects of the war in Ukraine may have modified the chance of success and should be considered when planning the next steps of a label launch.

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Literature cited

Bertschinger, L., Anderson, J.D., de Groot, N., Granatstein, D., Habib, R., Mullinix, K., Neilsen, D., Pomares García, F.,



- Weibel, F.P., and Zinati, G. (2004). Conclusions from the 1st Symposium on Sustainable Horticulture and a declaration for the 21st century. *Acta Hortic.* 638, 509–512 <https://doi.org/10.17660/ActaHortic.2004.638.65>.
- Calvert, G.M., Karnik, J., Mehler, L., Beckman, J., Morrissey, B., Sievert, J., Barrett, R., Lackovic, M., Mabee, L., Schwartz, A., et al. (2008). Acute pesticide poisoning among agricultural workers in the United States, 1998–2005. *Am. J. Ind. Med.* 51 (12), 883–898 <https://doi.org/10.1002/ajim.20623>. PubMed
- Darnhofer, I., Schneeberger, W., and Freyer, B. (2005). Converting or not converting to organic farming in Austria: farmer types and their rationale. *Agric. Human Values* 22 (1), 39–52 <https://doi.org/10.1007/s10460-004-7229-9>.
- Duggan, J. (2017). Implementing a metapopulation bass diffusion model using the R package deSolve. *R J.* 9 (1), 153 <https://doi.org/10.32614/RJ-2017-006>.
- Falconer, K. (2000). Farm-level constraints on agri-environmental scheme participation: a transactional perspective. *J. Rural Stud.* 16 (3), 379–394 [https://doi.org/10.1016/S0743-0167\(99\)00066-2](https://doi.org/10.1016/S0743-0167(99)00066-2).
- Flaten, O., Lien, G., Koesling, M., and Løes, A.K. (2010). Norwegian farmers ceasing certified organic production: characteristics and reasons. *J. Environ. Manage.* 91 (12), 2717–2726 <https://doi.org/10.1016/j.jenvman.2010.07.026>. PubMed
- Garming, H., Dirksmeyer, W., and Bork, L. (2018). Entwicklungen des Obstbaus in Deutschland von 2005 bis 2017: Obstarten, Anbauregionen, Betriebsstrukturen und Handel (No. 100). Thünen Working Paper.
- Geels, F.W., and Schot, J. (2007). Typology of sociotechnical transition pathways. *Res. Policy* 36 (3), 399–417 <https://doi.org/10.1016/j.respol.2007.01.003>.
- Howard, R.A. (1966). *Decision Analysis: Applied Decision Theory* (Stanford Research Institute).
- Li, S., and Kallas, Z. (2021). Meta-analysis of consumers' willingness to pay for sustainable food products. *Appetite* 163, 105239 <https://doi.org/10.1016/j.appet.2021.105239>. PubMed
- Luedeling, E., Goehring, L., Schiffers, K., Whitney, C., and Fernandez, E. (2021). Quantitative support of decision making under uncertainty. Contributed Package to the R Programming Language, Version 1.106. CRAN. <https://cran.r-project.org/web/packages/decisionSupport/index.html>
- Padel, S., and Foster, C. (2005). Exploring the gap between attitudes and behaviour: understanding why consumers buy or do not buy organic food. *Br. Food J.* 107 (8), 606–625 <https://doi.org/10.1108/00070700510611002>.
- Perneger, T.V., Courvoisier, D.S., Hudelson, P.M., and Gayet-Ageron, A. (2015). Sample size for pre-tests of questionnaires. *Qual. Life Res.* 24 (1), 147–151 <https://doi.org/10.1007/s11136-014-0752-2>. PubMed
- Reganold, J.P., Papendick, R.I., and Parr, J.F. (1990). Sustainable agriculture. *Sci. Am.* 262 (6), 112–121 <https://doi.org/10.1038/scientificamerican0690-112>.
- Rogers, E.M. (1962). How research can improve practice: a case study. *Theory Pract.* 1 (2), 89–93 <https://doi.org/10.1080/00405846209541785>.
- Rogers, E.M. (2003). *Diffusion of Innovations*, New York 1962.
- Schaer, O., Kourentzes, N., and Schaer, M.O. (2018). Package 'diffusion'.
- Schumpeter, J., and Backhaus, U. (2003). The theory of economic development. In Joseph Alois Schumpeter (Boston, MA: Springer), p.61–116.
- Stoneman, P., and Diederer, P. (1994). Technology diffusion and public policy. *Econ. J. (Lond.)* 104 (425), 918–930 <https://doi.org/10.2307/2234987>.
- Svanes, E., and Johnsen, F.M. (2019). Environmental life cycle assessment of production, processing, distribution and consumption of apples, sweet cherries and plums from conventional agriculture in Norway. *J. Clean. Prod.* 238, 117773 <https://doi.org/10.1016/j.jclepro.2019.117773>.
- Zikeli, S., Rembiałkowska, E., Załęcka, A., and Badowski, M. (2014). Organic farming and organic food quality: prospects and limitations. In *Sustainable Food Production Includes Human and Environmental Health. Issues in Agroecology – Present Status and Future Prospectus*, vol 3, W. Campbell, and S. López-Ortíz, eds. (Dordrecht: Springer).